



Time

Everyone of you must have seen clocks and watches in your home.

A clock or watch is a device which measures or tells the time.

Do you know how does a watch differ from a clock?

A clock is a wall-mounted instrument which tells the time.

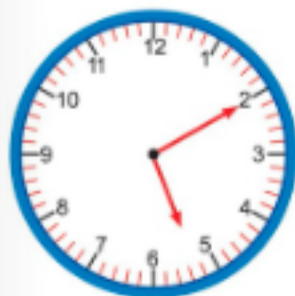
A watch is usually attached to a strap or band and worn around the wrist.



Both clocks and watches come in numerous shapes, designs and colours.

Digital and Analog Clocks/Watches

A digital clock or watch shows time using digits as shown here.

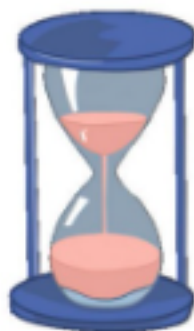


An analog or watch shows time by the movements of hour, minute and seconds hands.

In olden times, an hourglass or sand clock was used to tell the time.

It has two glass bulbs connected by a narrow neck that allows slow movement of sand from the upper bulb to lower bulb.

Try to find out how the hourglass works and discuss it with your classmates. You may even prepare a working model of the same using materials that you commonly see around you.





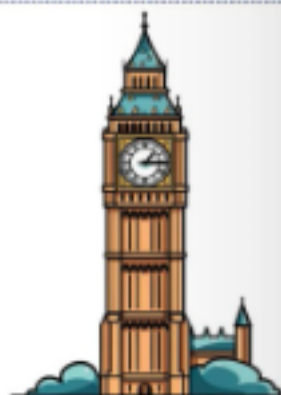
Did You Know?

Big Ben is the most famous clock tower in the world.

It is situated in the Palace of Westminster, London.

The tower has 11 floors.

The clock in this tower has been running since 1859 i.e. for about 165 years.



Reading Time

Look at the face or dial of clock shown here.

The dial of the clock is divided into 12 big divisions, marked as 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12.

The gap between every pair of consecutive numbers is divided into 5 equal small divisions. So, there are 60 small divisions on the whole dial. Each small division represents a minute.

The clock has two hands - a longer hand and a shorter hand.

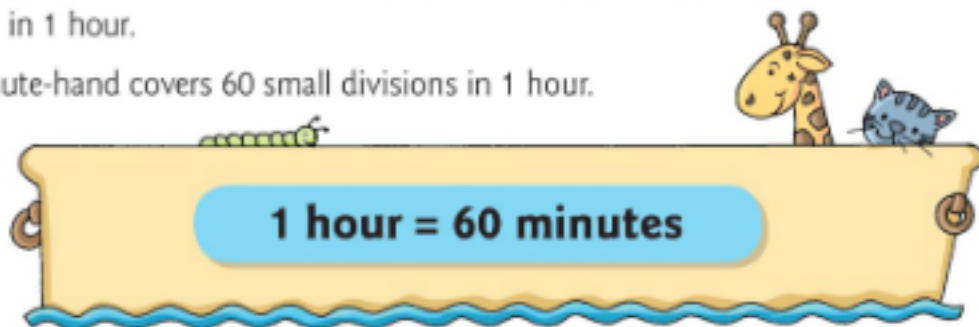
The longer hand is called the **minute hand**.

The shorter hand is called the **hour hand**.

The minute-hand moves from one small division to the next small division in 1 minute. It goes once round the dial in 1 hour.

Thus, the minute-hand covers 60 small divisions in 1 hour.

So,



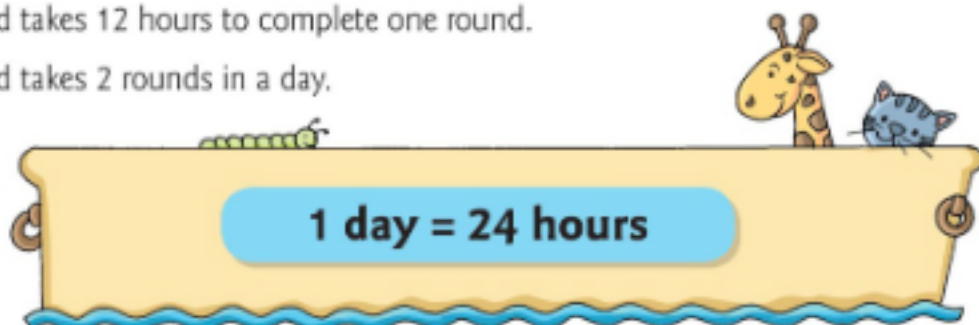
Note that the minute hand moves from one number to the next number in 5 minutes.

The hour hand moves from one number to the next number in 1 hour.

The hour hand takes 12 hours to complete one round.

The hour hand takes 2 rounds in a day.

So,



Reading Time From a Clock

To Read Time When the Minute Hand is at 12

Look at the clock shown on the right here.

The hour hand is at 7 and the minute hand is at 12.

We say that the time is **7 o'clock** or **7:00**.



In the clock shown on the left,
the hour hand is at 3 and the minute hand is at 12.

We say that the time is **3 o'clock** or **3:00**.

In the clock on the right,
the hour hand is at 11 and the minute hand is at 12.

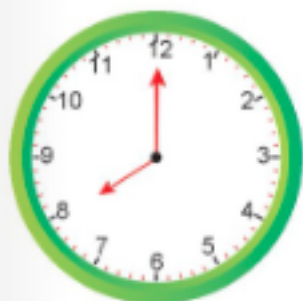
We say that the time is **11 o'clock** or **11:00**.



Exercise 56

Look at the clocks and write the correct time in each of the following:

1.



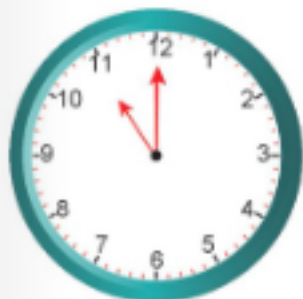
2.



3.



4.



5.



6.



The time is given and the minute hand is drawn on the clocks. Draw the hour hand to show the correct time.

7.



4 o'clock

8.



6 o'clock

9.



10:00

10. In the following table, the time is given. Write the positions of the minute hand and the hour hand.

	Time	Minute hand at	Hour hand at
(a)	5 o'clock		
(b)	9 o'clock		
(c)	3:00		
(d)	1:00		
(e)	12:00		

To Read Time When the Minute Hand is at any Number

When the minute hand moves from 12 to 1, it covers 5 small divisions. Similarly, upon reaching 2, it has covered 10 small divisions ; upon reaching 3, it has covered 15 small divisions; and so on.

Thus, minute hand at 1 shows $(1 \times 5) = 5$ minutes

minute hand at 2 shows $(2 \times 5) = 10$ minutes

minute hand at 3 shows $(3 \times 5) = 15$ minutes

minute hand at 4 shows $(4 \times 5) = 20$ minutes.

.....

minute hand at 8 shows $(8 \times 5) = 40$ minutes.

minute hand at 10 shows $(10 \times 5) = 50$ minutes.

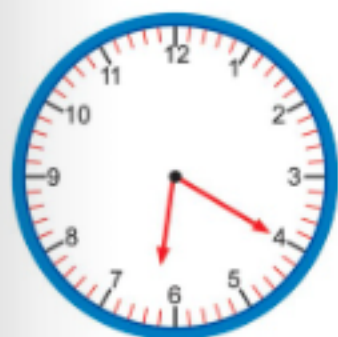
Look at the clock shown here.

The hour hand is beyond 8 and the minute hand is at 2, showing $(2 \times 5) = 10$ minutes.

So, the time is **8:10**.

We say that the time is **10 minutes past 8**.





Now, observe the clock shown on the left.

The hour hand is beyond 6 and the minute hand is at 4, showing $(4 \times 5) = 20$ minutes.

So, the time is 6:20.

We say that it is 20 minutes past 6.

Again, look at the clock on the right.

The hour hand is beyond 5 and the minute hand is at 10, showing $(10 \times 5) = 50$ minutes.

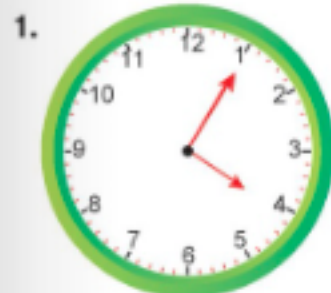
So, the time is 5:50.

We say that it is 50 minutes past 5.



Exercise 57

Look at each of the clocks given below and write down the correct time in numerals as well as in words.









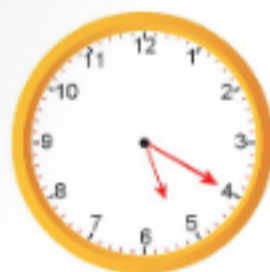




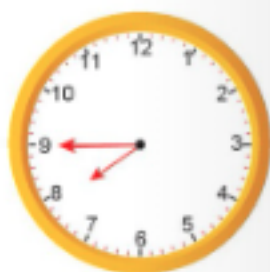
7.



8.



9.



To Read Time When the Minute Hand is at 6

Look at the clock shown here.

The hour hand is beyond 3.

The minute hand is at 6.

The minute hand at 6 shows 30 minutes.

So, we say that the time is **3:30** or **30 minutes past 3**.

Since 30 minutes shows half an hour, we also say that the time is **half past 3**.



In this clock, the hour hand is beyond 9.

The minute hand is at 6.

So, we say that the time is **9:30** or **30 minutes past 9** or **half past 9**.

In this clock,
the hour hand is beyond 12.
The minute hand is at 6.
So, we say that the time is **12:30** or **30 minutes past 12** or **half past 12**.



Exercise 58

Look at each clock and write the correct time shown by it in two ways.

1.



2.



3.



4.



5.

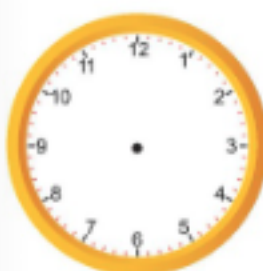


6.



Draw the hands of each of the following clocks to show the given time.

7.



8:30

8.



10:30

9.



Half past 6

To Read Time When the Minute Hand is at 3

Look at the clock shown here.

The hour hand is beyond 10.

The minute hand is at 3.

The minute hand at 3 shows 15 minutes.

So, we say that time is **10:15** or **15 minutes past 10**.

Since 15 minutes make **quarter** of an hour, so

we can also say that the time is **quarter past 10**.



In this clock,

the hour hand is beyond 2.

The minute hand is at 3.

So, we say that the time is **2:15** or

15 minutes past 2 or **quarter past 2**.

In this clock,

the hour hand is beyond 6.

The minute hand is at 3.

So, we say that the time is **6:15** or **15 minutes past 6**

or **quarter past 6**.





Exercise 59

Look at each clock and write the correct time shown by it in two ways.

1.



2.



3.



4.



5.

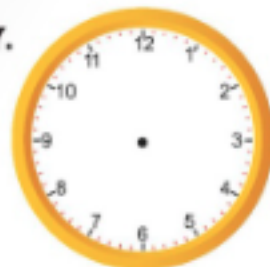


6.



Draw the hands of each of the following clocks to show the given time.

7.



6:15

8.



10:15

9.



Quarter past 4

To Read Time When the Minute Hand is at 9

Look at the clock shown here.

The hour hand is beyond 2. The minute hand is at 9.

The minute hand at 9 shows 45 minutes.

So, we say that the time is **2:45** or **45 minutes past 2**.

Clearly, it is 15 minutes less to be 3 o'clock.

Thus, we may also say that the time is **quarter to 3**.





In this clock,
the hour hand is beyond 11.
The minute hand is at 9.
So, we say that the time is **11:45** or **45 minutes past 11**
or **quarter to 12**.

In this clock,
the hour hand is beyond 5.

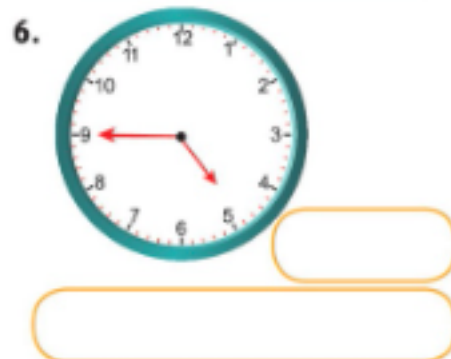
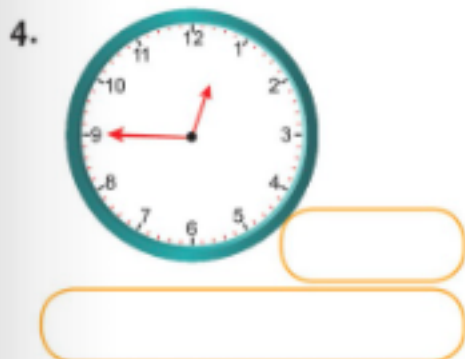
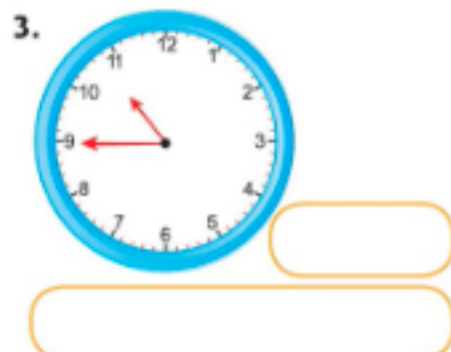
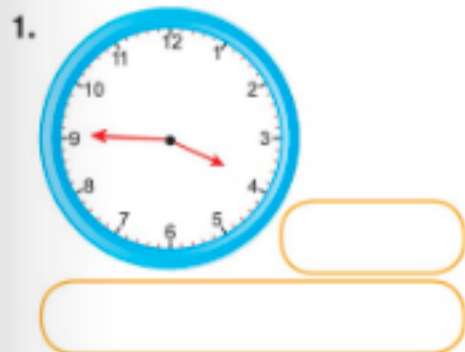
The minute hand is at 9.

So, we say that the time is **5:45** or **45 minutes past 5**
or **quarter to 6**.

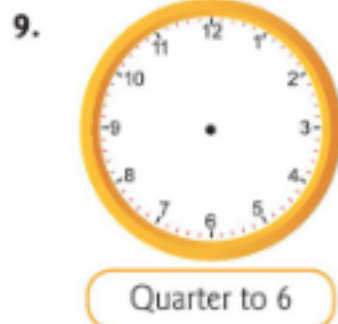
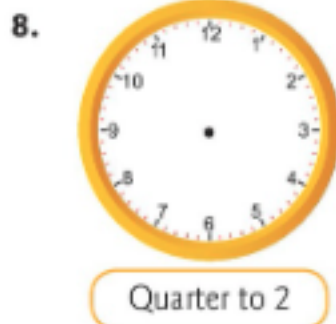
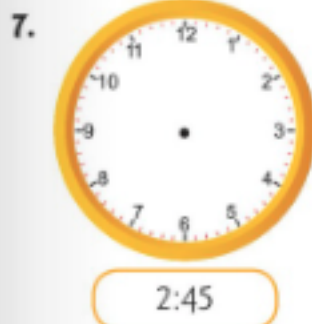


Exercise 60

Look at each clock and write the correct time shown by it in two ways.



Draw the hands of the clocks showing the given time.



Daily Routine

WAKE UP TIME

The minute hand is at

The hour hand is beyond

∴ The time is

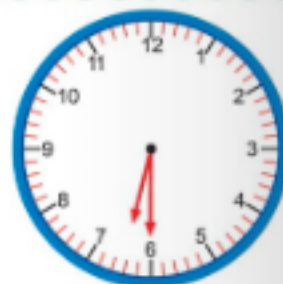


BATHING TIME

The minute hand is at

The hour hand is beyond

∴ The time is

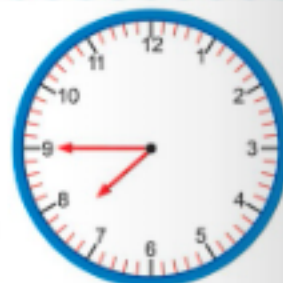


BREAKFAST TIME

The minute hand is at

The hour hand is beyond

∴ The time is



SCHOOL GOING TIME

The minute hand is at

The hour hand is at

∴ The time is

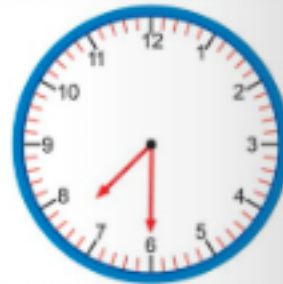


SCHOOL STARTS

The minute hand is at

The hour hand is beyond

∴ The time is



LUNCH TIME AT SCHOOL

The minute hand is at

The hour hand is beyond

∴ The time is



SCHOOL LEAVING TIME

The minute hand is at

The hour hand is at

∴ The time is



PLAYING TIME

The minute hand is at

The hour hand is at

∴ The time is



STUDY TIME

The minute hand is at

The hour hand is beyond

∴ The time is



T.V. WATCHING TIME

The minute hand is at

The hour hand is beyond

∴ The time is



DINNER TIME

The minute hand is at

The hour hand is beyond

∴ The time is



SLEEPING TIME

The minute hand is at

The hour hand is at

∴ The time is





Activity Time



Note the time taken by you for various daily activities from morning to night, on a certain day and prepare a chart as shown below:

Activity	Starting Time	Finishing Time	Time Taken
Bathing			
Eating lunch			
Doing homework			

After preparing the above chart, you must be acquainted with the correct usage of units of time—minutes and hours.

Using this knowledge, tick (✓) the approximate time required for the given job / activity:

1. Make a phone call 5 minutes / 5 hours
2. Tie your shoelaces 2 minutes / 2 hours
3. Watch a movie 20 minutes / 2 hours
4. Play with friends 10 minutes / 1 hour
5. Eat your breakfast 15 minutes / 2 hours
6. Sleep at night 8 minutes / 8 hours



Things to Remember

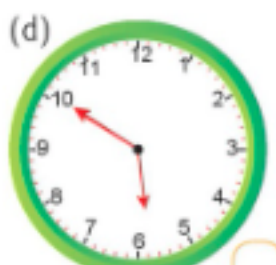
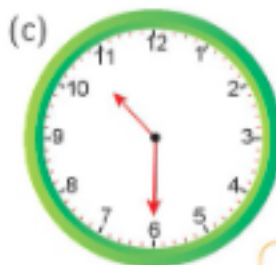
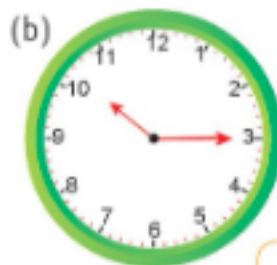
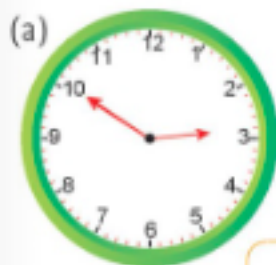
1. In a clock, the larger hand is the minute hand and the shorter hand is the hour hand.
2. 1 hour = 60 minutes
3. 1 day = 24 hours
4. The minute hand moves from one number to the next number in 5 minutes.
5. The hour hand moves from one number to the next number in 1 hour.
6. The minute hand completes 24 rounds while the hour hand completes 2 rounds of the clock in a day.

**QUESTION BAG 1****(Objective Type Questions)****Tick (✓) the correct answer.**

1. What is the correct time shown in the clock?

(a) Quarter to 4 ☐(b) Half past 3 ☐(c) 3 o' clock ☐(d) Quarter past 3 ☐

2. Which of the following clocks shows the time as 10:30?



3. When the minute hand is at 10, it has covered minutes.

(a) 30 ☐(b) 40 ☐(c) 50 ☐(d) 60 ☐

4. When the minute hand is at 8, it means it has minutes to complete the hour.

(a) 10 ☐(b) 15 ☐(c) 20 ☐(d) 30 ☐

5. 10 minutes to 12 can be read as

(a) 11:40 ☐(b) 11:50 ☐(c) 12:10 ☐(d) 12:50 ☐

6. How many rounds does the minute hand of a clock complete in a day?

(a) 2 ☐(b) 12 ☐(c) 24 ☐(d) None of these ☐

7. Rohan went to play at the time shown in clock A and came back at the time shown in clock B. How long did Rohan play ?

**Clock A****Clock B**(a) 20 minutes ☐(b) 25 minutes ☐(c) 30 minutes ☐(d) 60 minutes ☐

8. Which of the following is not correctly matched?

(a) 6:30 – Half past 6 ☐(b) 11:15 – Quarter to 11 ☐(c) 9:45 – Quarter to 10 ☐(d) 3:10 – 10 minutes past ☐

9. How many hours does the hour hand complete in one round of the clock?
 (a) 1 ☐ (b) 2 ☐ (c) 12 ☐ (d) 24 ☐
10. At what time both the hands of a clock are exactly opposite to each other?
 (a) 6:00 ☐ (b) 12:30 ☐ (c) 3:45 ☐ (d) 9:15 ☐
11. The minute hand takes minutes to move from 10 to 12.
 (a) 5 ☐ (b) 10 ☐ (c) 15 ☐ (d) 20 ☐

QUESTION BAG 2

1. Complete the following table.

Minute hand on	1	2	3	4	5	6	7	8	9	10	11	12
Minutes passed	5	10										

2. Fill in the blanks.

- (a) The minute hand takes minutes to move from 4 to 7.
- (b) The hour hand takes hours to move from 1 to 4.
- (c) At both the hands will be at 12.
- (d) The hand of the clock is the minute hand while the hand is the hour hand.
- (e) When the time is a quarter past an hour, the minute hand is always at
- (f) 2:15 is also read as past
- (g) A quarter past 6 is also read as
- (h) At a quarter to an hour, the minute hand is always at
- (i) When it is a quarter to an hour, the minute hand has covered minutes.
- (j) A quarter to 5 is also read as
- (k) In one day, the hour hand goes round the clock.

3. State whether each of the following statements is true or false.

- (a) Quarter to 7 is written as 6:45.
- (b) Half past 2 is written as 2:15.
- (c) The hour hand moves faster than the minute hand.
- (d) The hour hand moves from 1 to 3 in 10 minutes.
- (e) At 9:30, the minute hand is between 9 and 10.